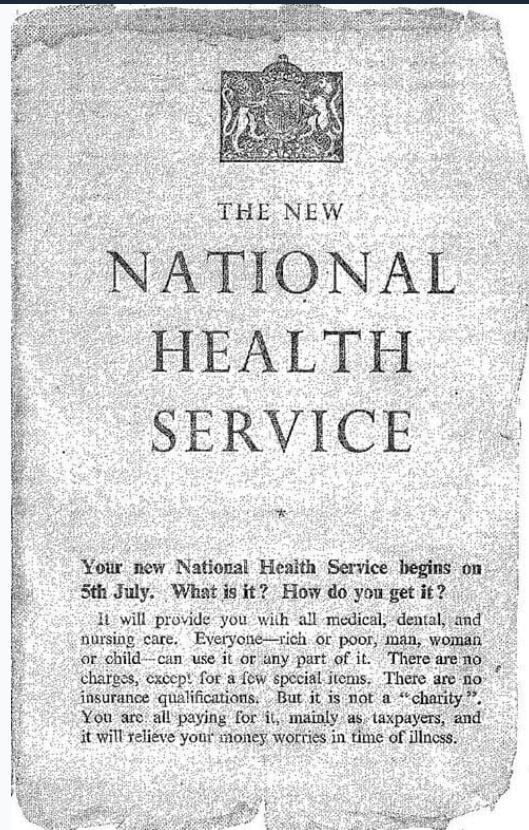




WHY HAS MEDICINE ADVANCED MORE IN THE 20TH CENTURY THAN IN ALL PREVIOUS HISTORY COMBINED?

Paper 1: Medicine Through Time with the Western Front – Modern

Scholar: _____

Class: _____

Edexcel Specification: c1900–present

- **1. Causes of disease:** Lifestyle factors, genetics, and the role of the NHS.
- **2. Prevention:** Immunisation programmes, screening, and public health campaigns.
- **3. Treatment:** Fleming and penicillin; blood transfusions; transplants; radiotherapy.
- **4. Modern Britain:** The founding of the NHS (1948) and its ongoing impact.

PROGRESS & ASSESSMENT TRACKER

TARGET GRADE: _____

Grade	9-1 GCSE Grading Scale		
Lesson / Assessment Title	Effort	Level	Teacher Comments
L1: KT4.1: How have modern discoveries changed our understanding of illness?			
L2: KT4.2: How have prevention and treatment advanced in the modern era?			
L3: KT4.3: How was Penicillin discovered and mass-produced?			
L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?			
Final Unit Grade:			

L1: KT4.1: How have modern discoveries changed our understanding of illness?

LEARNING OBJECTIVES

☐ Explain how the discovery of DNA and the Human Genome Project transformed the understanding of genetic and hereditary illnesses.

☐ Analyze the influence of lifestyle choices on modern health and the impact of technology on medical diagnosis.

Do Now Activity

Score: / 10

What disease did Edward Jenner vaccinate against in 1796?

What did Edward Jenner observe that led to his vaccine?

Who discovered the cause of cholera in 1854?

Which Roman physician developed the Theory of Opposites?

What was the Royal Society?

Who introduced antiseptic surgery using carbolic acid in 1865?

What did Harvey prove wrong about Galen's theories of blood?

Who published the Germ Theory in 1861?

What invention helped spread new medical ideas during the Renaissance?

What was 'transference'?

Vocabulary Check

The Odd One Out: Circle ONE term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

DNA | Rosalind Franklin | Human Genome Project | Biopsy | Electron Microscope | Inherited Diseases

KNOWLEDGE CHECK (Q1)

How did Watson and Crick's discovery of the double-helix structure of DNA help scientists search for the causes of hereditary diseases?

- ☐ A. It provided a complete blueprint of human biology, allowing scientists to identify specific faulty genes responsible for conditions like cystic fibrosis.
- ☐ B. It proved that hereditary diseases were caused by viruses.
- ☐ C. It allowed scientists to create antibiotics to kill hereditary diseases.
- ☐ D. It proved that hereditary diseases were caused by miasma.

Q2. Identify who discovered the double-helix structure of DNA in 1953.

Q3. Describe the goal of the Human Genome Project.

Q4. Explain why Germ Theory could not explain all illnesses.

Q5. Evaluate the impact of the discovery of DNA on modern medicine.

Pair & Share Activity

Q6. Prompt: Which modern development has had a more significant impact on our understanding of the causes of illness: the discovery of genetic factors (DNA and hereditary disease) or the scientific identification of lifestyle factors (smoking, diet, and alcohol)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how mapping the genome allows women to identify risks for breast cancer, versus how smoking is proven to be the biggest cause of preventable diseases in the world, causing heart disease and lung cancer).

Your Notes:

16-MARK ESSAY PLANNING MATRIX (2×2 GRID)

Edexcel Paper 1 • Section B (Q5/Q6)

Question: 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the 20th century.' How far do you agree? Explain your answer.

Stimulus: Watson and Crick • Lifestyle factors

✓ Factors Supporting Statement (Agree)

- Watson and Crick's 1953 discovery of the double-helix DNA structure (using Rosalind Franklin's X-ray crystallography) unlocked the fundamental genetic code governing human biology.
- The Human Genome Project (completed 2003) enabled scientists to map all 20,000–25,000 human genes, pinpointing hereditary causes for over 4,000 genetic conditions including cystic fibrosis and breast cancer (BRCA1/2).

⇌ Counter-Factors / Other Factors (Disagree)

- Epidemiological research into **lifestyle factors** had a far more immediate impact on public health: Doll and Hill's 1950 study proved the causal link between smoking and lung cancer, alongside dietary and alcohol health links.
- Advanced physical diagnostic imaging technology (X-rays from 1895, ultrasound from the 1950s, CT scans from the 1970s, and MRI from the 1980s) allowed doctors to detect internal tumors and organ failure without genetic analysis.

Level 4 Judgment Criteria: Weigh the relative importance: Did the discovery of DNA provide the supreme theoretical breakthrough in etiology, or did lifestyle epidemiology and diagnostic scanning technology deliver greater practical impact for patients?

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal blue lines across its entire width. The lines are thin and consistent in color, set against a plain white background. There are no margins, text, or other markings present on the page.

Q8. Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness.

[12 marks • 15 mins]

[illegible]

[illegible]

Q9. Explain one way in which ideas about the cause of illness in the modern period (c1900-present) were different from ideas in the period c1700-c1900.

[4 marks • 5 mins]

Q10. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c.1900-present.' How far do you agree? Explain your answer.

You may use the following in your answer:

- Watson and Crick
- Lifestyle factors

You must also use information of your own.

General Notes

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L2: KT4.2: How have prevention and treatment advanced in the modern era?

LEARNING OBJECTIVES

- ☐ Explain how scientific and technological developments led to the creation of magic bullets and antibiotics.
- ☐ Analyze the impact of the National Health Service (NHS) on the provision of and access to medical care.

Do Now Activity

Score: / 10

What structure did Watson and Crick discover in 1953?

What is the function of DNA in relation to disease?

Name one modern lifestyle factor that causes disease.

Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery?

Why did the mapping of the human genome not lead to immediate cures?

What did Harvey prove wrong about Galen's theories of blood?

Who published the Germ Theory in 1861?

Who introduced antiseptic surgery using carbolic acid in 1865?

Who wrote 'On the Fabric of the Human Body' (1543)?

What was the significance of the Public Health Act of 1875?

Vocabulary Check

The Golden Sentence: Choose TWO terms from the word bank. Write ONE grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (because, although, or consequently):

Magic Bullet | Salvarsan 606 | Prontosil | Antibiotics | National Health Service (NHS) | Beveridge Report

KNOWLEDGE CHECK (Q1)

What is the key scientific difference between an antiseptic like carbolic acid and a magic bullet like Salvarsan 606?

- ☐ A. Antiseptics are used on the outside of the body to prevent infection, whereas magic bullets are injected into the body to target specific microbes.
- ☐ B. Antiseptics are created from microorganisms, whereas magic bullets are synthetic chemicals.
- ☐ C. Antiseptics are used to cure viruses, whereas magic bullets cure bacteria.
- ☐ D. There is no difference; they are both the same thing.

Q2. Identify the first 'Magic Bullet' discovered in 1909.

Q3. Describe how Magic Bullets work.

Q4. Explain how diagnosing illness changed in the 20th century.

Q5. To what extent did the discovery of Magic Bullets revolutionize treatment?

Pair & Share Activity

Q6. Prompt: Which development was the most significant turning point in twentieth-century healthcare: the scientific breakthrough of antibiotics, or the political creation of the National Health Service (NHS)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how penicillin cured previously fatal infections, or how the NHS removed the fear of poverty by providing free healthcare for millions).

Your Notes:

Q7. Explain why the government took a more active role in preventing disease in the period c1900-present.

[12 marks • 15 mins]

[illegible]

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Q8. Explain one way in which ideas about the prevention of illness in the modern period (c1900-present) were different from the medieval period (c1250-c1500).

[4 marks • 5 mins]

Q9. Explain why there was rapid progress in disease prevention in the period c1900-present.

[12 marks • 15 mins]

You may use the following in your answer:

- Government vaccination campaigns
- Legislation

You must also use information of your own.

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[illegible]

General Notes

L3: KT4.3: How was Penicillin discovered and mass-produced?

LEARNING OBJECTIVES

- ☐ Contrast the roles of Alexander Fleming with those of Howard Florey and Ernst Chain in the penicillin breakthrough.
- ☐ Analyze how the factors of government intervention, science and technology, and war collaborated to enable the mass production of penicillin.

Do Now Activity

Score: / 10

What is a 'magic bullet'?

Name the first magic bullet, discovered by Paul Ehrlich in 1909.

What was established in Britain in 1948 to provide free medical care?

Who was the Minister of Health responsible for establishing the NHS in 1948?

Name the second magic bullet, discovered by Gerhard Domagk in 1932 to treat strep infections.

What structure did Watson and Crick discover in 1953?

Who published the Germ Theory in 1861?

Who introduced antiseptic surgery using carbolic acid in 1865?

Name one modern lifestyle factor that causes disease.

Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery?

Vocabulary Check

Conceptual Classification: Categorise the terms from the word bank into the two historical boxes below:

Penicillin | Alexander Fleming | Florey and Chain | Culture Medium | Mass Production | Antibiotic Resistance

Power, Governance & Warfare

Economy, Trade & Society

Q1. The Discovery and Mass Production of Penicillin

GCSE TWO-SIDED ARGUMENT

GCSE Analytical Assessment: Evaluate the miraculous antibacterial breakthrough of penicillin against its industrial challenges and modern limitations.

▲ Antibiotic Miracle (Change) - Alexander Fleming observed in 1928 that penicillium notatum mold destroyed staphylococcus bacteria colonies on an open culture plate. - Howard Florey and Ernst Chain at Oxford (1938–1941) successfully isolated and purified penicillin, proving it cured deadly bacterial infections in mice and human trials. - US industrial scale and War Production Board funding produced 2.3 million doses in 1944, saving thousands of Allied soldiers on D-Day and eliminating pneumonia/gangrene. <i>Starter: “On the one hand, the development of penicillin was the greatest therapeutic breakthrough of the 20th century because...”</i>	■ Industrial Fragility & Antibiotic Resistance - Fleming abandoned his discovery in 1929 because he could not extract enough pure penicillin; without Florey, Chain, and US funding, it would have remained a laboratory footnote. - Oxford trials showed how fragile the supply was: policeman Albert Alexander showed rapid recovery but died when the team ran out of penicillin. - Penicillin was completely useless against viral infections like influenza, measles, or the common cold. - Modern over-prescription in humans and livestock has bred dangerous antibiotic-resistant superbugs like MRSA. <i>Counter: “However, the penicillin breakthrough had profound limitations because...”</i>
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GCSE Paragraph Builder: Write a balanced GCSE exam paragraph evaluating how far penicillin was an individual discovery versus an industrial collaboration with modern vulnerabilities.

Connectives: Although Fleming made the initial serendipitous discovery... • Crucially, transforming this into a viable drug required... • Furthermore, in contemporary medicine... • Ultimately...

Q2. Describe how Penicillin was discovered.

KNOWLEDGE CHECK (Q3)

What was the main scientific and practical limitation that caused Fleming to abandon his work on penicillin in 1931?

- ☐ A. He realized that penicillin was highly toxic to humans.
- ☐ B. He could not secure funding from the government because it was peacetime.

- ☐ C. He lacked the chemical expertise and facilities to extract and purify penicillin in large enough quantities for human trials.
- ☐ D. He was drafted into the army to fight in World War II.

Q4. Explain the roles of Florey and Chain in the development of Penicillin.

Q5. Evaluate the role of World War II in the mass production of Penicillin.

Pair & Share Activity

Q6. Prompt: Who deserves the most credit for the penicillin breakthrough: Alexander Fleming for his accidental discovery in 1928, or Howard Florey and Ernst Chain for their scientific development and therapeutic isolation of the drug in the 1940s?

Think: Jot down your choice and one piece of evidence from the sources (e.g., Fleming recognizing and publishing the mould's bacteria-killing properties in 1928, or the Oxford team actually turning it into a usable drug and proving its clinical effectiveness on humans).

Your Notes:

16-MARK ESSAY PLANNING MATRIX (2×2 GRID)

Edexcel Paper 1 • Section B (Q5/Q6)

Question: 'Alexander Fleming was the most important individual in the development of penicillin.' How far do you agree?
Explain your answer.

Stimulus: Howard Florey and Ernst Chain • The Second World War

✓ Factors Supporting Statement (Agree)

- Alexander Fleming made the initial serendipitous breakthrough in 1928, recognizing that *Penicillium notatum* mould produced an antibacterial substance that dissolved *Staphylococcus* bacteria.
- Fleming published his findings in the *British Journal of Experimental Pathology* (1929); without his keen observation of the contaminated petri dish, penicillin would never have been discovered.

⇌ Counter-Factors / Other Factors (Disagree)

- Fleming failed to chemically purify penicillin and abandoned research in 1931; Howard Florey and Ernst Chain's Oxford team revived the work in 1938, developing purification techniques and proving efficacy on mice (1940) and human patient Albert Alexander (1941).
- The US government and American pharmaceutical corporations provided millions in funding during WWII, utilizing deep-tank corn-steep liquor fermentation to mass-produce 2.3 million doses by D-Day (June 1944), transforming mortality rates.

Level 4 Judgment Criteria: Weigh the relative importance: Was Fleming's original accidental observation the paramount contribution, or did Florey and Chain's scientific purification and wartime industrial mass-production matter more for clinical survival?

Q7. Explain why there was rapid progress in the development of penicillin in the 1940s.

[12 marks • 15 mins]

[illegible]

[illegible]

Q8. Explain one way in which the treatment of illness in the modern period (c1900-present) was different from the treatment of illness in the period c1700-c1900.

[4 marks • 5 mins]

Q9. 'Alexander Fleming was the most important individual in the development of penicillin.'
How far do you agree? Explain your answer.

[20 marks • 25 mins]

You may use the following in your answer:

- Staphylococcus
- Florey and Chain

You must also use information of your own.

- Staphylococcus
- Florey and Chain

You must also use information of your own.

[illegible]

[illegible]

[illegible]

General Notes

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L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

LEARNING OBJECTIVES

- ☐ Explain how modern science and technology have transformed the diagnosis and treatment of lung cancer since c1900.
- ☐ Analyze the effectiveness and changing focus of government legislation to prevent smoking-related lung cancer.

Do Now Activity

Score: / 10

Who discovered Penicillin by accident in 1928?

Which two scientists at Oxford developed Penicillin into a usable drug?

What major global event drove the mass production of Penicillin?

What is a 'magic bullet'?

What was established in Britain in 1948 to provide free medical care?

Who was the Minister of Health responsible for establishing the NHS in 1948?

What are the four humours?

Who wrote 'On the Fabric of the Human Body' (1543)?

Why is lung cancer so difficult to treat effectively compared to other diseases?

Which landmark 1950 epidemiological study proved the link between smoking and lung cancer?

Vocabulary Check

Spot the Deliberate Error: Read the statement below. One historical fact or vocabulary concept has been deliberately falsified. Underline the error and explain the accurate historical reality below:

Carcinogen | Bronchoscopy | CT Scanner | Radiotherapy | Immunotherapy | Nanny State

Challenge: Write ONE statement containing a deliberate historical misconception using **Carcinogen** or **Bronchoscopy**. Swap with a partner to identify and correct the error:

Q1. Identify one method used to diagnose Lung Cancer.

Q2. Describe the aggressive treatments used for Lung Cancer.

Q3. Explain why the UK government has focused heavily on preventing Lung Cancer.

KNOWLEDGE CHECK (Q4)

What is the key clinical advantage of using a CT scan over a traditional chest X-ray when diagnosing a lung cancer patient?

- ☐ A. A CT scan uses harmless sound waves instead of radiation.
- ☐ B. A CT scan can be performed at home by the patient.

- ☐ C. A CT scan creates highly detailed, 3D cross-sectional images of the lungs, allowing doctors to detect much smaller tumors than a 2D X-ray.
- ☐ D. A CT scan can actively cure the lung cancer by destroying the tumor.

Q5. Evaluate the effectiveness of government legislation in reducing smoking.

Pair & Share Activity

Q6. Prompt: Which category of modern government action has been the most significant turning point in the prevention of lung cancer: directly warning and educating current smokers (such as plain packaging and graphic warning labels), preventing young people from starting (such as raising the legal age of sale from 16 to 18), or protecting non-smokers from second-hand smoke (such as the 2007 indoor smoking ban)?

Think: Jot down your choice and one piece of evidence from your knowledge (for instance, the 2007 ban significantly shifting social attitudes about where smoking is acceptable, or how 1950s research first proved the definitive link between tobacco and cancer).

Your Notes:

Q7. Explain how science and technology have improved the diagnosis and treatment of lung cancer.

[12 marks • 15 mins]

[illegible]

[illegible]

Q8. Explain one way in which hospital care in the modern period (c1900-present) was different from hospital care in the period c1700-c1900.

[4 marks • 5 mins]

Q9. Explain why the government took action to improve public health in the period c1900-present.

[12 marks • 15 mins]

You may use the following in your answer:

- Lung cancer
- The NHS

You must also use information of your own.

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General Notes

[illegible]

End of Unit Reflection & Pupil Voice

Unit: Paper 1: Medicine Through Time

Instructions: Now that you have completed your final assessment, take 10 minutes to reflect on your learning across this entire unit. Be honest—this is your chance to use your **Pupil Voice** to help your teacher plan future lessons.

1. What Went Well (WWW)

What topic or skill did you find most engaging or easiest to understand in this unit?

2. Even Better If (EBI)

What did you find most challenging? Are there any concepts you are still confused about?

3. Effort & Target Setting

Circle your effort level for this unit: **1 (Low)** **2** **3** **4** **5 (Excellent)**

What is one specific target you want to set for yourself in the next unit?

Teacher Feedback & Coaching

(To be completed during live marking / feedback lessons)

Factors Overview: Modern

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	